

Swiss Institute of Banking and Finance



University of St.Gallen

10. Market Efficiency and Behavioral (Corporate) Finance

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3,130/4,130 Corporate Finance

Course outline

1. General Information
2. Risk and Return
3. Portfolio Theory and CAPM
4. Corporate Valuation
5. Cost of Capital
6. Corporate Financing and the Issuance of Securities
7. Capital Structure
6. Risk Management
9. Conflicts of Interest and Corporate Governance
- 10. Market Efficiency and Behavioral (Corporate) Finance**
11. Empirical Research in Corporate Finance

After this lecture you will have understood...

- why stock prices are often assumed to follow a **random walk** and what this means for stock market predictability,
- the concept of **market efficiency** and its implications for price formation on financial markets,
- that researchers have documented **empirical evidence** for and against market efficiency, and
- how **behavioral finance** challenges human rationality and explains observed inefficiencies with (distorted) risk attitudes and beliefs about probabilities.

Are markets efficient? – Yes!

- When the Challenger space shuttle crashed on 28 January 1986, the share price of the firm responsible for manufacturing the faulty part dropped significantly more than the ones of the three other contractors.
- This happened within the first 20 minutes after the crash, while it took an official panel months to determine which component was at fault (Maloney & Mulherin, 2003, p. 453).
- Another example: Roll (1984) shows that the price of frozen concentrated orange juice is a better predictor of the central Florida weather, with 98% if the US production being in central Florida, than any other US weather service.



Are markets efficient? – No!

The New York Times

HOPE IN THE LAB: A special report.; A Cautious Awe Greet Drugs That Eradicate Tumors in Mice

 Give this article



By Gina Kolata

May 3, 1998

Within a year, if all goes well, the first cancer patient will be injected with two new drugs that can eradicate any type of cancer, with no obvious side effects and no drug resistance -- in mice.

Some cancer researchers say the drugs are the most exciting treatment that they have ever seen. But then they temper their enthusiasm with caution, noting that the history of cancer treatments is full of high expectations followed by dashed hopes when drugs with remarkable effects in animals are tested in people.

Are markets efficient? – No! (II)

- On May 3, 1998, the New York Times published an article on an apparently ground-breaking new strain in cancer research by a company called EntreMed Inc.
- Following this news, the company's opening share price surged to over seven times its Friday closing value of around USD 12 on the following Monday, with prices stabilizing throughout a day with notably increased trading volumes at a level above USD 50.
- While the New York Times article was placed prominently on the cover page, the same information had already been published over five months earlier in the scientific as well as the popular press.
- The market appears to have undervalued this information at its first release in November 1997, while overvaluing it in May 1998, before finding an equilibrium price of USD 25-30 (Huberman and Regev, 2001).

A startling discovery: Price changes are random

- Do stock prices follow any particular pattern? Can we predict the return of a stock for the next day or month?
- In 1953, Maurice Kendall, a British statistician, presented a controversial paper on the behavior of stock and commodity prices to the Royal Statistical Society.
- *Kendall* had expected to find regular price cycles...
- ...but to his surprise they did not seem to exist: Each series appeared to be

“a ‘wandering’ one, almost as if once a week the Demon of Chance drew a random number... and added it to the current price to determine the next week’s price.”

A startling discovery: Price changes are random (II)

- Kendall's idea was not new: It had been proposed in 1900 in an almost forgotten doctoral thesis ("Théorie de la Speculation") by French mathematician Louis Bachelier (1870-1946).
- Bachelier's accompanying development of the mathematical theory of random processes anticipated by five years *Einstein's* famous work on the random **Brownian motion** of colliding gas molecules.
- In spite of the original work and enormous contribution to the mathematics and economics fields, Bachelier's academic career was not very successful and only in 1927 he found a permanent full professor position at the University of Besançon.



Definition: Random walk

- A random walk is a random process in a discrete period $t=1,2, \dots$, i.e., a sequence of random states $\tilde{X}_t$ that are connected to each other as follows:

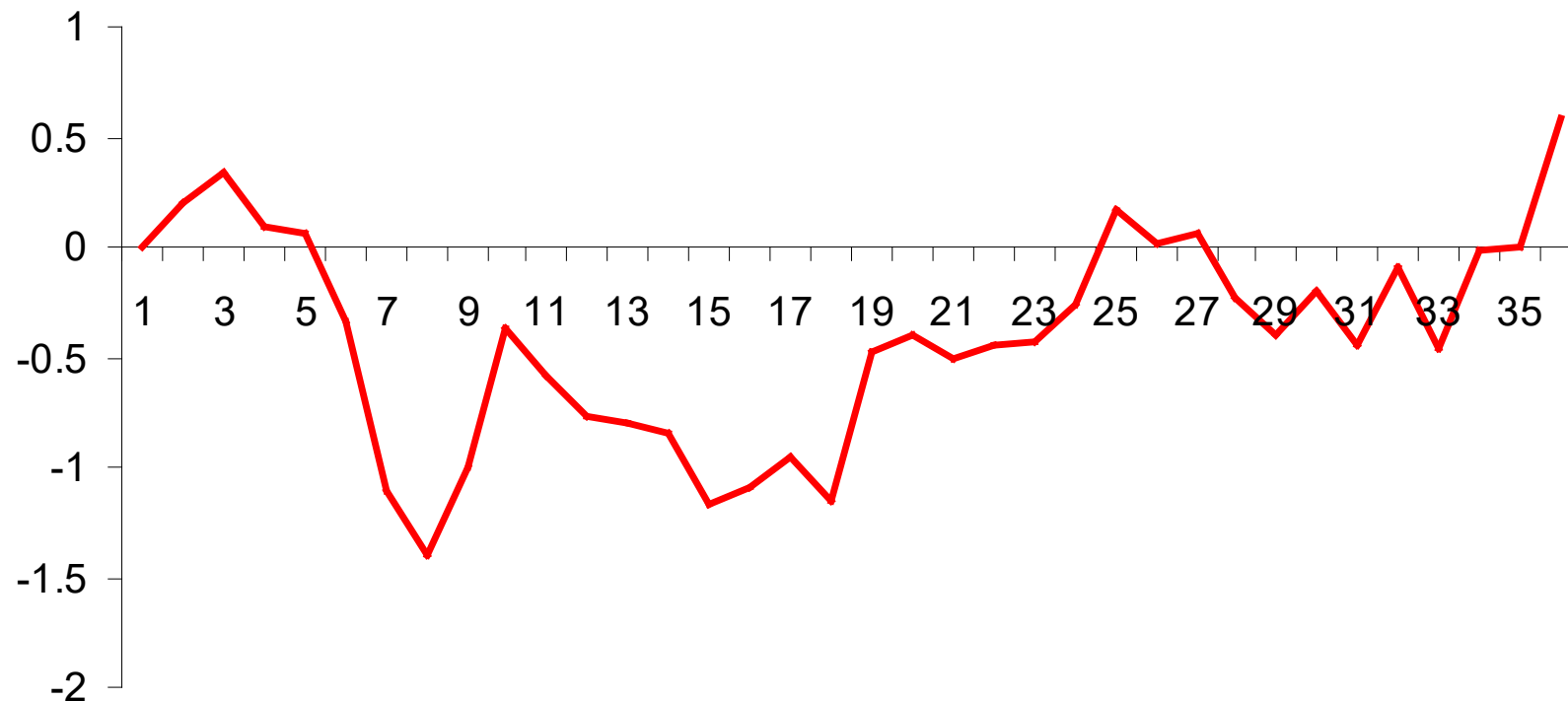
$$\tilde{X}_t = \tilde{X}_{t-1} + \tilde{r}_t$$

- The random increments $\tilde{r}_1, \tilde{r}_2, \dots$ are independent and identically distributed (iid):

$$Cov[\tilde{r}_{t+1}, \tilde{r}_t] = 0$$

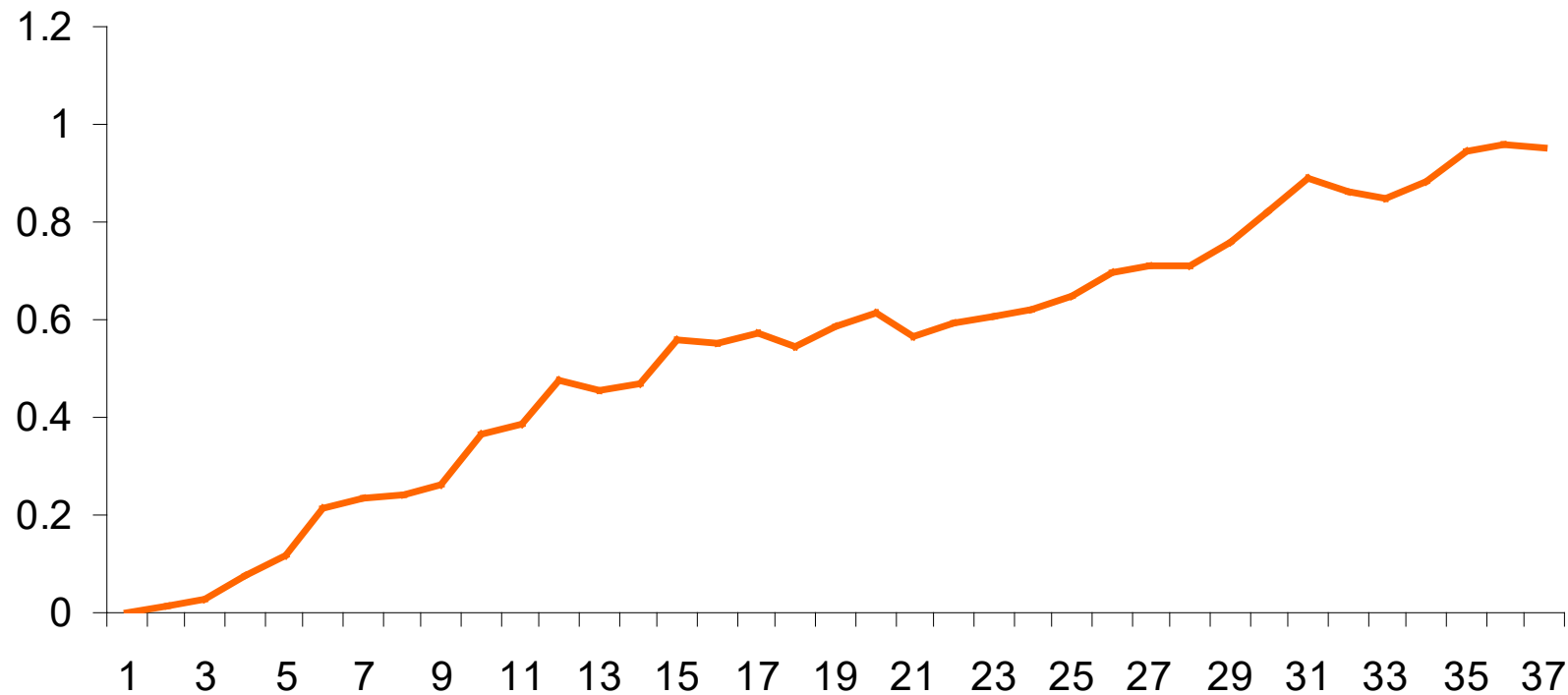
Random walk – Example

- A random walk without “drift” ($E[\tilde{r}_t] = 0$):

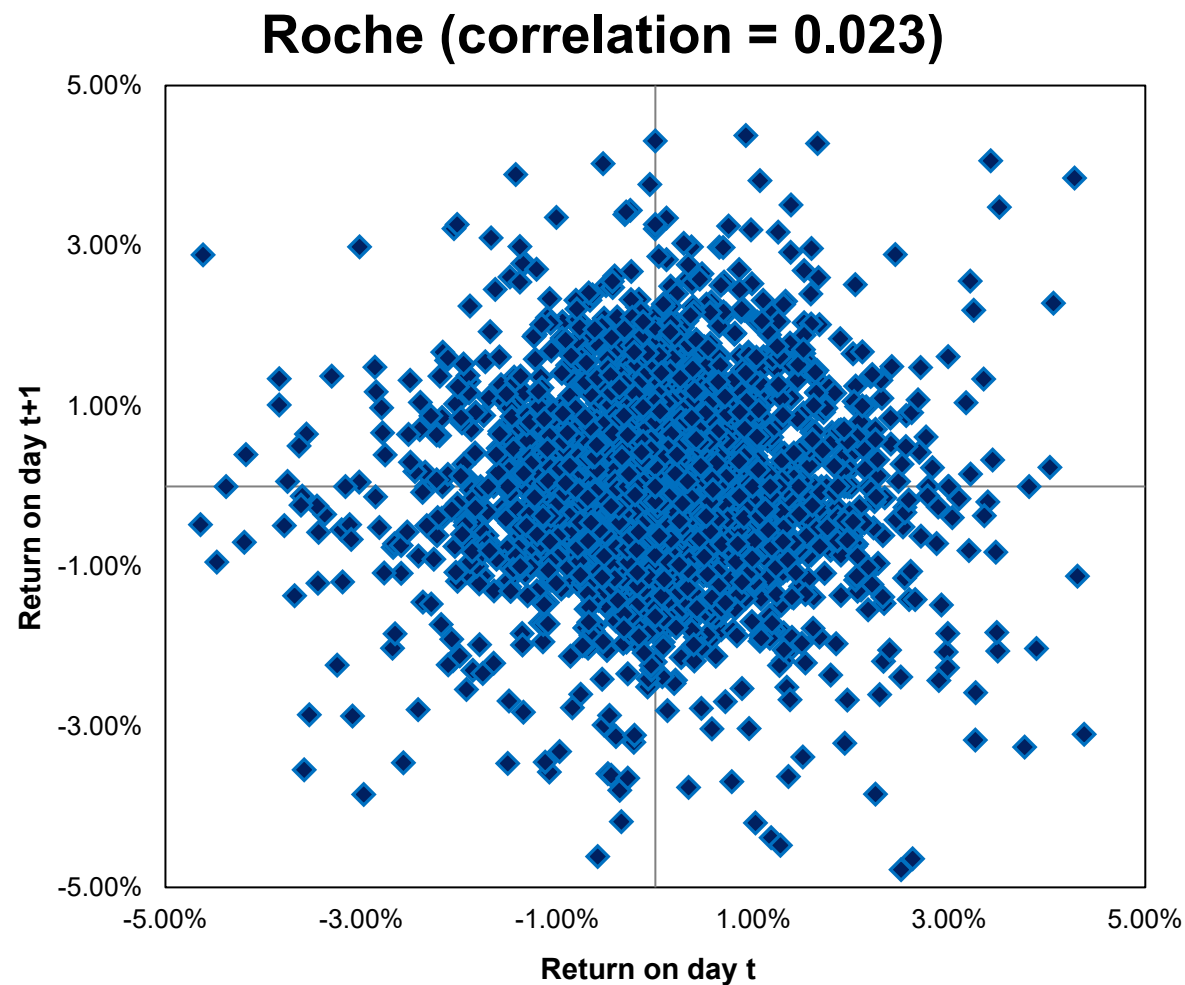


Random walk – Example (II)

- A random walk with “drift” ($E[\tilde{r}_t] > 0$):



The reality: Price changes are random



Historical daily returns 1/2009 – 12/2018
Source: Datastream

There is no free lunch...

- If past price changes could predict future price changes, one could easily make a profit. But there is **no free lunch**.
- Randomness is not a sign of irrationality: It is consistent with rational market participants.
- As soon as new information is released, market participants process this information and prices adjust accordingly: This adjustment process is very fast as only the fastest traders will make a profit from the new information.
- Speed matters: High frequency traders have rented apartments very close to the NYSE to save a few milliseconds when routing their trades to the exchange. In recent years, stock exchanges have started to rent out computer space on their premises.

There is no free lunch... (II)

- Academic research looks into questions related to trading speed:



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Every Cloud Has a Silver Lining: Fast Trading, Microwave Connectivity, and Trading Costs

ANDRIY SHKILKO and KONSTANTIN SOKOLOV*

ABSTRACT

Modern markets are characterized by speed differentials, with some traders being fractions of a second faster than others. Theoretical models suggest that such differentials may have both positive and negative effects on liquidity and gains from trade. We examine these effects by studying a series of exogenous weather episodes that temporarily remove the speed advantages of the fastest traders by disrupting their microwave networks. The disruptions are associated with lower adverse selection and lower trading costs. In additional analysis, we show that the long-term removal of speed differentials results in similar effects and also increases gains from trade.

Market efficiency

Share prices in an efficient market:

- Prices in financial markets immediately and correctly reflect all publicly available information – as well as everything that can be derived from it.
- Unpredictability: As price changes reflect the random flow of new information, price changes must also occur randomly.
- Random Walk: Prices change erratically as new – and value relevant – information arrives.
- Eugene Fama, one of the most influential academics in finance and leading contributor of the “Chicago School”, proposed three forms of market efficiency.

Three forms of market efficiency

weak-form efficiency

A market is weak-form efficient if it is not possible to achieve systematically superior investment performance (on average, across segments, and periods of market occurrences) on the basis of the analysis of historical prices and returns → they are well-known and incorporated in today's prices.

semi-strong-form efficiency

A market is semi-strong-form efficient if it is not possible to achieve systematically superior investment performance (on average, across segments, and periods of market occurrences) on the basis of all publicly available information.

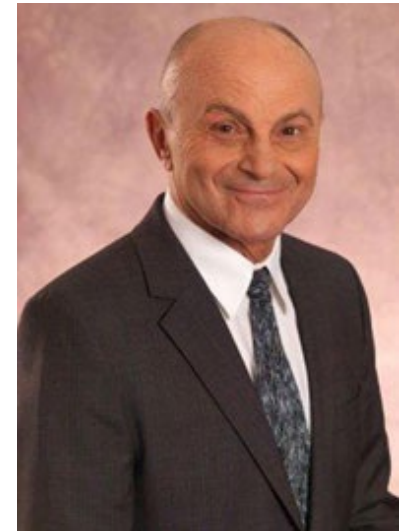
strong-form efficiency

A market is strong-form efficient if it is not possible to achieve systematically superior investment performance (on average, across segments, and periods of market occurrences) on the basis of any information, public or private. → Hence, even insider information cannot be exploited!

Eugene F. Fama

Eugene Francis “Gene” Fama (1939*) is widely recognized as the "father of modern finance." He is the joint recipient of the 2013 Nobel Prize in Economic Sciences with Lars Peter Hansen of the University of Chicago and Robert J. Shiller of Yale University. Fama's financial research is well known in both academia and the investment community. He is strongly identified with research on markets, particularly with regard to the efficient market hypothesis. Through his research he has brought an empirical and scientific rigor to the field of investment management, transforming the way finance is viewed and conducted.

He earned a bachelor's degree from Tufts University in 1960, followed by an MBA and PhD from the University of Chicago Booth School of Business in 1964. His doctoral supervisors were Nobel prize winner Merton Miller and Harry Roberts, but Benoit Mandelbrot was also an important influence. He has spent all of his teaching career at the University of Chicago.



How does market efficiency work?: Takeovers

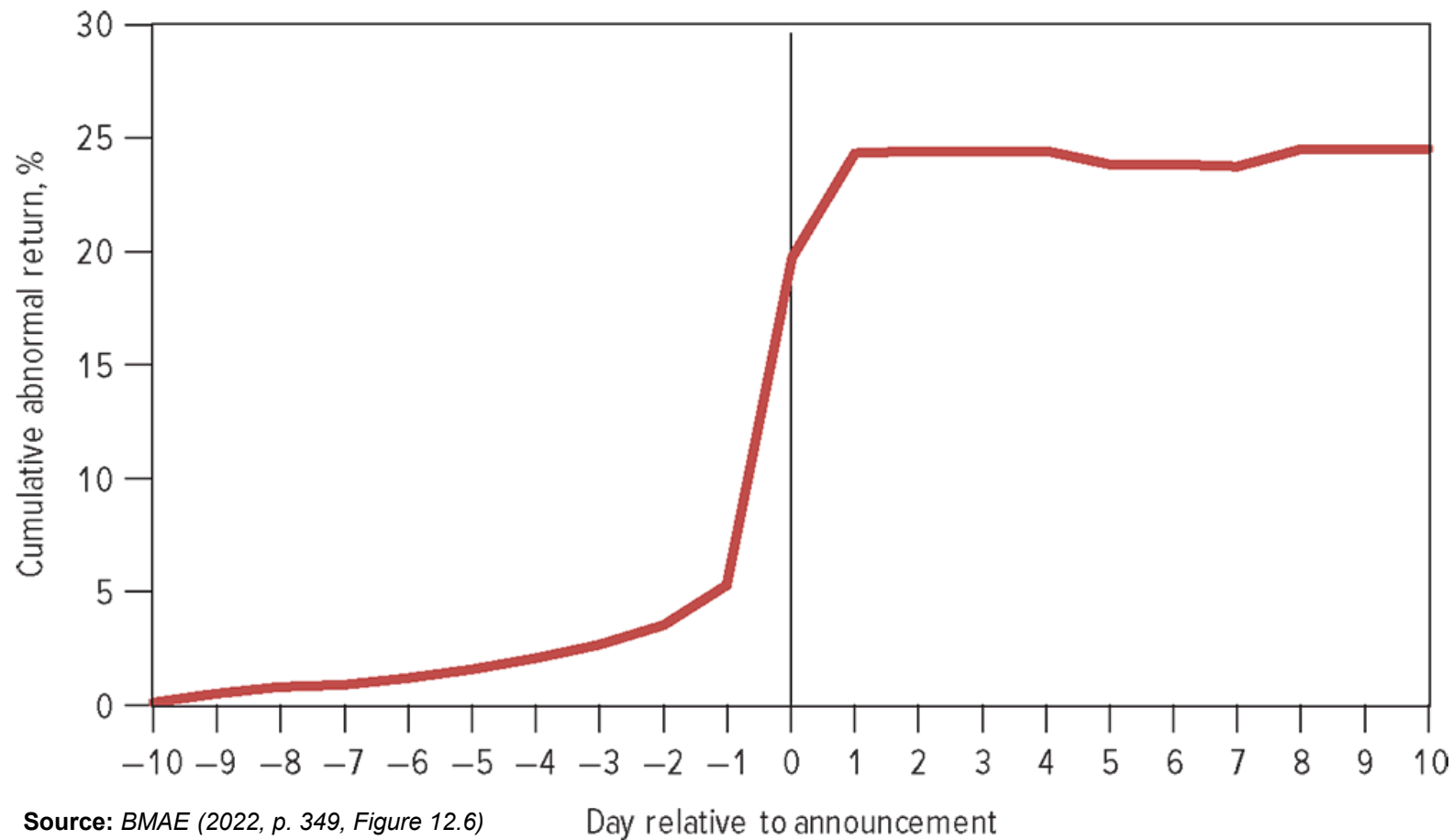
- To empirically test whether market efficiency holds, researchers have analyzed how fast security prices respond to different types of information, such as earnings, dividend, or takeover announcements.
- For takeovers this test of market efficiency has practical implications as well.
- **Example:** AB Technologies decides to buy a smaller publicly-traded competitor in Switzerland (CB Data), and hires an investment bank to structure the process. Nobody except senior management at AB Technologies and the mandated bank know about the up-coming takeover attempt.

How does market efficiency work?: Takeovers (II)

- **Example continued:** After analyzing the company, the investment bank suggests to offer a premium of 20% over the current market price of CHF 50 in order to convince investors to tender their shares to the bidder.
- It is legally forbidden to disclose this tender price before the offer is made available to the public (**insider trading**).
- Thus, the bank and the regulator will monitor the movement of the stock price before the takeover announcement date. Any drift towards the offer price before the announcement is public could indicate information leakage and insider trading.

How does market efficiency work?: Takeovers (III)

Price reaction to takeover announcements (8,668 U.S. targets, 1975-2019):



Event study approach

- As the name suggests, an event study transforms various “events” from calendar time to **event time**, i.e., each announcement takes place on day $t = 0$.
- To isolate the effect of an event on the stock return, we have to look at abnormal returns rather than raw returns.
- Often this is done with the **market model**:

$$AR_i = r^{observed} - r_f - \beta_i(r_m - r_f)$$

- These ARs are then averaged across events to obtain an estimate of the announcement effect on the day of announcement.
- Often, event dates are measured imprecisely and cumulative abnormal returns (CARs) over a few days are investigated.

How academics understand market efficiency

- When academics refer to “market efficiency”, they usually have one of two things in mind:
 - **There is no free lunch:** There are no arbitrage opportunities, i.e., riskless profits without any investment are not possible.
 - **Prices are equal to value:** The market price of a security is equal to its fundamental value (e.g., the discounted stream of future dividends).
- Notice that these **definitions are not identical!** For instance, the price of a stock might not be equal to its fundamental value, but arbitrageurs cannot act on this price difference (due to transaction costs, short-sale constraints,...)

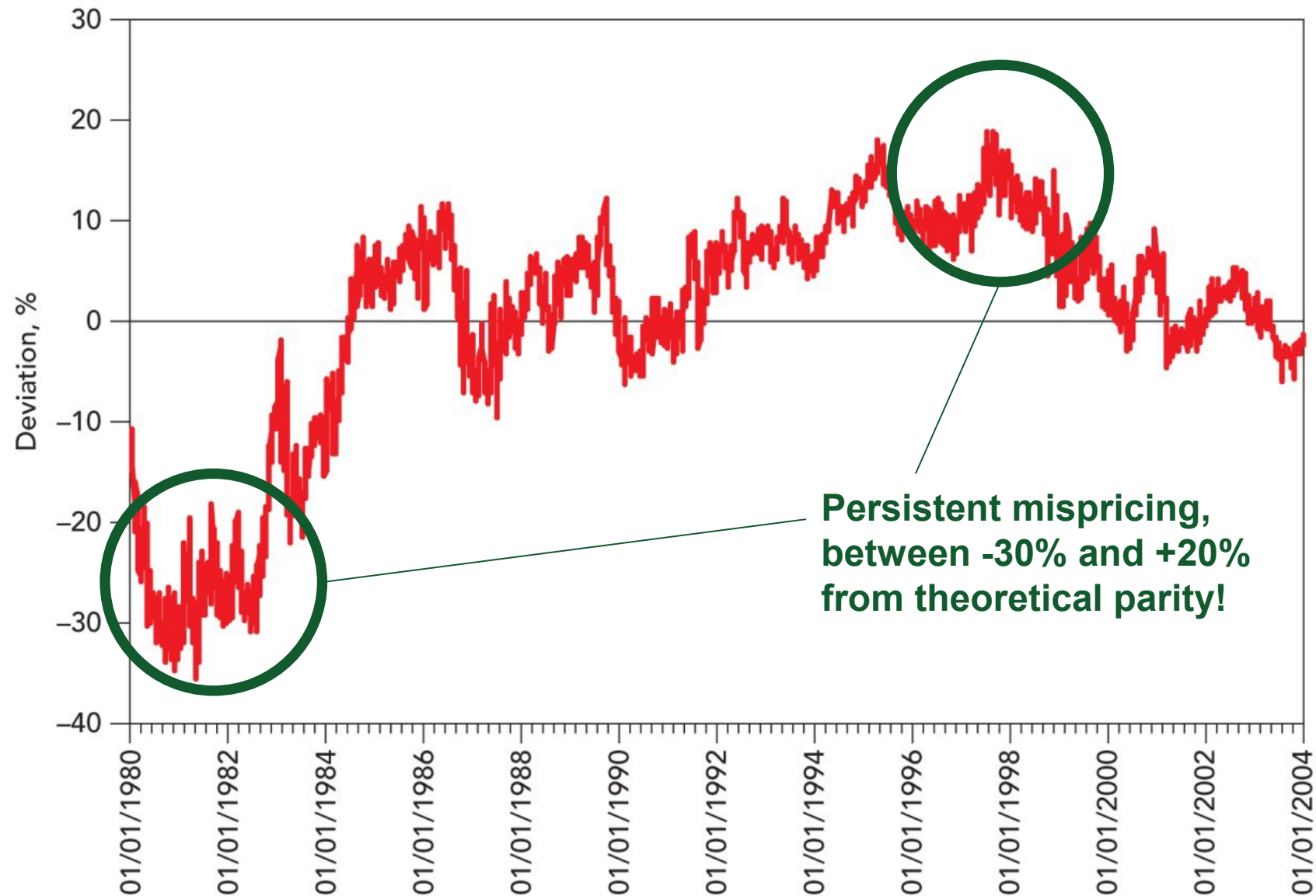
Challenging market efficiency

- To **challenge market efficiency**, find examples where either the assumption of **no free lunch** or of **price equal to value** is violated.
- If you find two assets with identical cash flows which do not sell for the same price, one market efficiency assumption is violated.
- Here are two examples of challenges to market efficiency which academics have documented:
 1. Siamese twin companies (Royal Dutch Shell)
 2. Equity carve outs (3COM/ Palm)

Siamese twin companies: Royal Dutch Shell

- In 1907 Royal Dutch and Shell merged business: There is one parent company and the cash flows are split: 60% going to Royal Dutch (trading in Amsterdam) and 40% to Shell (trading in London).
- If market efficiency holds, Royal Dutch should be valued at 1.5x the price of Shell because of the 60/40 cash flow split.
- In reality, the ratio of both prices was persistently and significantly different from this theoretical price parity.
 - This violates the assumption that two assets with identical cash flows should have the same price!

Siamese twin companies: Royal Dutch Shell (II)



Source: BMAE (2022, p. 358, Figure 12.8)

Equity carve outs: 3Com

- Equity carve outs are capital market transactions in which a company decides to sell off part of its business in the form of an IPO.
- In 2000, 3Com decided to sell 5% of Palm (its subsidiary) to the public, while retaining the other 95%. When Palm went public, each 3Com shareholder indirectly owned 1.5 shares of Palm (because of the retained shares).
- After the IPO, Palm was trading at approx. 95 USD, so that each 3Com shareholder's stake in Palm was worth around 143 USD (95×1.5).
- But, 3Com also had approx. 10 USD cash per outstanding share, and other profitable business lines.

Equity carve outs: 3Com (II)

- In total, one 3Com share was theoretically worth approx. 178 USD.
- But 3Com was trading at 82 USD per share!
- This implies that the value of all other assets from 3Com was -61 USD per share. 3Com's stake in Palm alone was worth 51 USD billion, but 3Com's market value was only 28 USD billion!
- This apparent miss-pricing lasted for over a year.
- The limited free float of Palm shares (only 5%) might have inflated the share price, as most shares were also held by retail investors. This introduced short sale constraints as mostly institutional shareholder lend stocks for short selling.

Tip of the iceberg or rare anomalies?

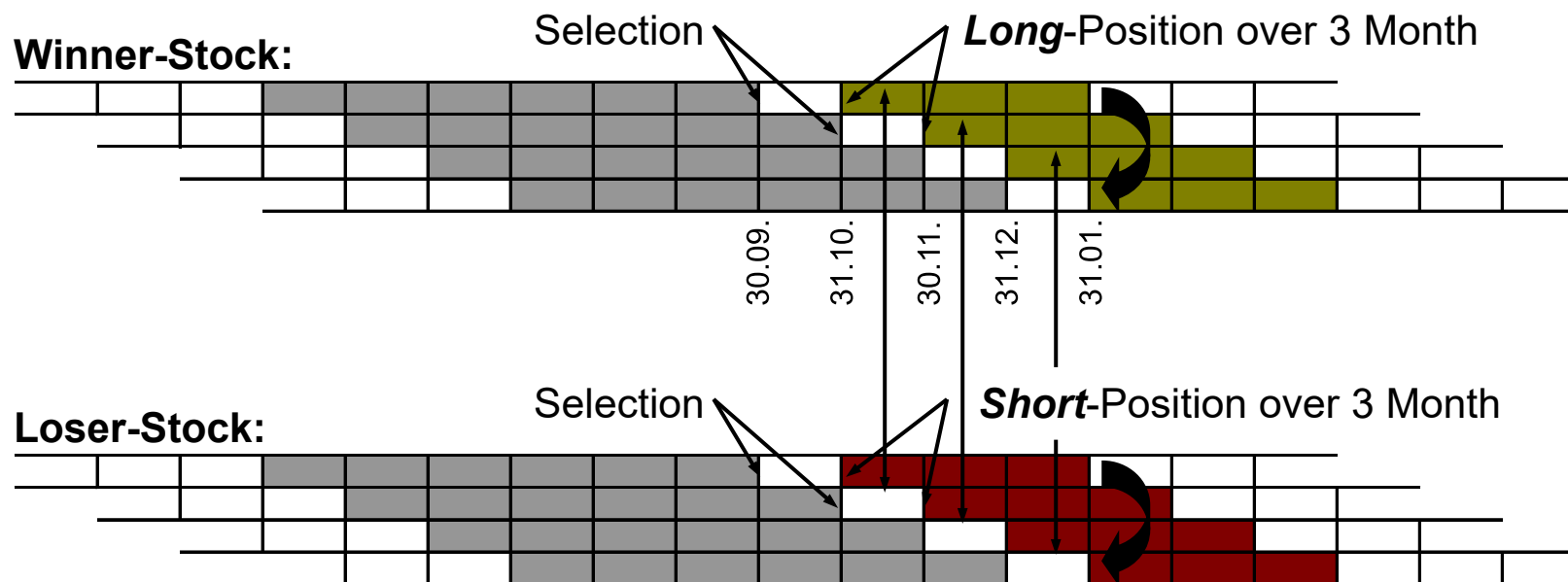
- Promoters of the “Chicago School” of efficient markets might argue that these examples are rare anomalies that do not shatter the overall efficiency of capital markets.
- As a matter of fact, the two examples may not necessarily offer arbitrage opportunities because they may be too risky or too expensive to exploit.
- Are such violations of market efficiency only the tip of the iceberg or hard to find?
- From a practical point of view, one might not care whether a price is different from its fundamental value if one cannot act on it (i.e., change the mispricing, like in the Royal Dutch Shell case).

A challenge even to weak form efficiency: Momentum

- Momentum: A simple strategy that buys past “winner” stocks and sells short past “loser” stocks.
- Rey and Schmid (2007) analyze “feasible” (= cost efficient) momentum strategies in the Swiss stock market.
- Stock universe: Swiss Market Index (SMI), 1994-2004
- To assess the real performance of the strategy: The index must be exactly replicated and include all stocks that were ever included. Ex-ante rules need to be applied to stocks that disappear (because of takeover, delisting, bankruptcy) or drop out of the SMI.

A challenge even to weak form efficiency: Momentum (II)

- Strategy setup (example: $J = 6$; $K = 3$):



- After three months, the long and short pair is sold/closed and a new investment is made.
- Different rebalancing horizons across the K investment strands are tested.

A challenge even to weak form efficiency: Momentum (III)

Table 1 Annualized average returns to arbitrage strategies, zero-cost arbitrage strategies and winner and loser portfolios

<i>J</i>	<i>K</i>	Arbitrage strategy 1994–2004 (1)	Zero-cost arb. strat. 1994–2004 (2)	Arbitrage strategy 1994–1999:6 (3)	Zero-cost arb. strat. 1994–1999:6 (4)	Arbitrage strategy 1999:7–2004 (5)	Zero-cost arb. strat. 1999:7–2004 (6)
3	3	9.19%* (0.0726)	13.09% (0.2485)	4.17% (0.3190)	4.43% (0.4154)	14.45%* (0.0804)	22.41% (0.2488)
3	6	14.34%*** (0.0066)	18.91%** (0.0407)	5.47% (0.2148)	7.63% (0.1487)	23.95%*** (0.0058)	31.27%* (0.0521)
3	12	12.08%*** (0.0034)	13.46%** (0.0409)	12.87%*** (0.0014)	13.70% (0.1491)	11.29%* (0.0616)	13.22% (0.2081)
6	3	43.79%*** (0.0000)	50.87%*** (0.0005)	12.89%* (0.0586)	17.76%** (0.0141)	83.15%*** (0.0000)	92.32%*** (0.0007)
6	6	26.53%*** (0.0000)	39.17%*** (0.0025)	14.54%*** (0.0076)	17.15%** (0.0327)	39.76%*** (0.0000)	64.92%*** (0.0052)
6	12	21.88%*** (0.0000)	21.94%** (0.0236)	16.78%*** (0.0000)	17.56% (0.1103)	27.20%*** (0.0002)	26.47%* (0.0918)
12	3	15.72%** (0.0128)	29.24%** (0.0488)	13.06%* (0.0594)	19.52% (0.1644)	18.44%* (0.0504)	39.69% (0.1097)
12	6	20.10%*** (0.0002)	38.26%** (0.0153)	19.19%*** (0.0012)	26.04%* (0.0870)	21.02%* (0.0134)	51.56%* (0.0642)
12	12	19.75%*** (0.0000)	22.35% (0.1269)	16.15%*** (0.0002)	14.30% (0.2814)	23.47%*** (0.0006)	30.92% (0.2437)

A challenge even to weak form efficiency: Momentum (IV)

- Is this result specific to Switzerland (or the time period)?
- Ammann, Moellenbeck, and Schmid (2011) test similar strategies on the S&P 100 index from 1982 to 2009.
- However, they only buy winners and go short in the S&P 100.
- Ann. returns to $J = 6$ strategies:
 - $K = 3$: 18.24%
 - $K = 6$: 15.96%
 - $K = 12$: 12.96%

Figure 1b: Formation Period $J = 6$ months



Behavioral finance

- In short, behavioral finance tries to explain why prices can depart from fundamental values.
- Underlying assumption: People are not always rational.
- Behavioral finance usually relies on insights from psychology and attempts to explain empirically observed patterns which are inconsistent with rational investors and efficient markets.
- „Behavioural finance is the study of the influence of psychology on the behaviour of financial practitioners and the subsequent effect on markets.“ Sewell (2005)
- „I think of behavioral finance as simply, open-minded finance‘.“ Thaler (1993)

Behavioral finance – Attitude about risk

- Consider the following lottery: You can choose between a sure gain of CHF240, or you can gamble with a 25% chance to win CHF1000, and 75% to get nothing.
- A rational and risk-neutral person would compute the expected values (CHF240 vs. CHF250), and choose the second option.
- But people are risk averse. In a seminal experiment, conducted by Daniel Kahneman and Amos Tversky, 84% of the respondents chose the first option.
 - Generally, people are risk-averse when it comes to choices involving gains.

Behavioral finance – Attitude about risk (II)

- Let's change the framing of the lottery: Now, you can choose between a sure **loss** of CHF750, or you can gamble with a 25% chance of losing nothing, and a 75% probability of losing CHF1000.
- A rational and risk-neutral person would be indifferent between these choices.
- When the same respondents were asked, 87% chose the riskier alternative, which could result in even higher losses.
 - Generally, people are risk-seeking when it comes to choices involving losses!

Behavioral finance – Attitude about risk (III)

- A rational person would not be influenced by the framing of the lottery (i.e., whether it involves gains or losses).
- People often exhibit inconsistent behavior depending on the framing of the lottery: They are risk-averse regarding gains (selling their stocks after they have increased by 10%) but risk-seeking regarding losses (not selling their stocks even if they have lost 20% in value): **Disposition effect**.

Behavioral finance – Acceptance in finance

Daniel Kahneman
Princeton University
Nobel prize, 2002



"for having integrated insights from psychological research into economic science, especially concerning human judgment and decision-making under uncertainty"

Vernon Smith
George Mason University
Nobel prize, 2002



"for having established laboratory experiments as a tool in empirical economic analysis, especially in the study of alternative market mechanisms"

Behavioral finance – Beliefs about probabilities

- People make all sorts of systematic biases when assessing the likelihood of uncertain events or when interpreting data.
- In this lecture we will have a closer look at three such behavioral biases:
 1. Anchoring
 2. Overconfidence
 3. Ambiguity aversion

Behavioral finance – Anchoring

- Numerous studies have shown that people are influenced by **(numerical) anchors**.
- For instance, Kahneman and Tversky have asked a group of people to estimate the product of $1*2*3*4*5*6*7*8$ and another group to estimate the product of $8*7*6*5*4*3*2*1$.
 - It turned out that the first number shown served as an anchor. The first group's average estimate was 512, the second group's 2,250 (significantly different, but still far of from the correct answer, 40,320).
 - The same behavior can be observed when **trading stocks!** The purchase price serves as an anchor when deciding whether to sell the stocks in the future (which is irrational).

Behavioral finance – Overconfidence

- Generally, people are overconfident when assessing probabilities about uncertain events or when comparing themselves to their peers (e.g., everyone is sure to be the best car driver around).
- Overconfidence often leads investors to hold themselves responsible for good performance, but uncontrollable circumstances and bad luck for bad performance.
- The result is a distorted, overly optimistic self-assessment that may trigger excessive trading and high trading costs.

Behavioral finance – Ambiguity aversion

Ellsberg Paradox (1961):

Urn 1



50 blue balls
50 red balls

100 balls

Urn 2



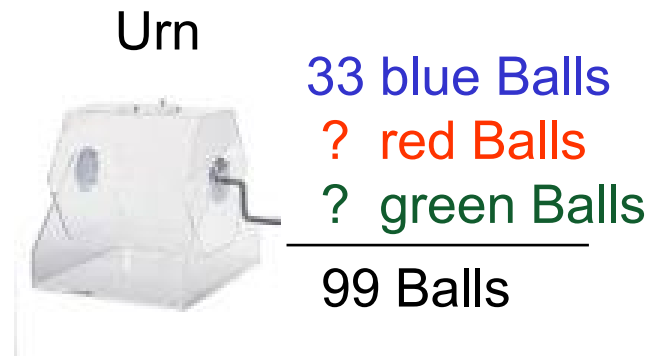
? blue balls
? red balls

100 balls

Which lottery would you prefer, urn 1 or urn 2?

Behavioral finance – Ambiguity aversion (II)

Ellsberg Paradox II (1961):



You may choose one of two lotteries in the following two games:

- A: 100 CHF if “blue”
- B: 100 CHF if “red”

- C: 100 CHF if “blue or green”
- D: 100 CHF if “red or green”

Which lottery would you prefer between A and B? And between C and D?

Behavioral finance: Individual trading behavior

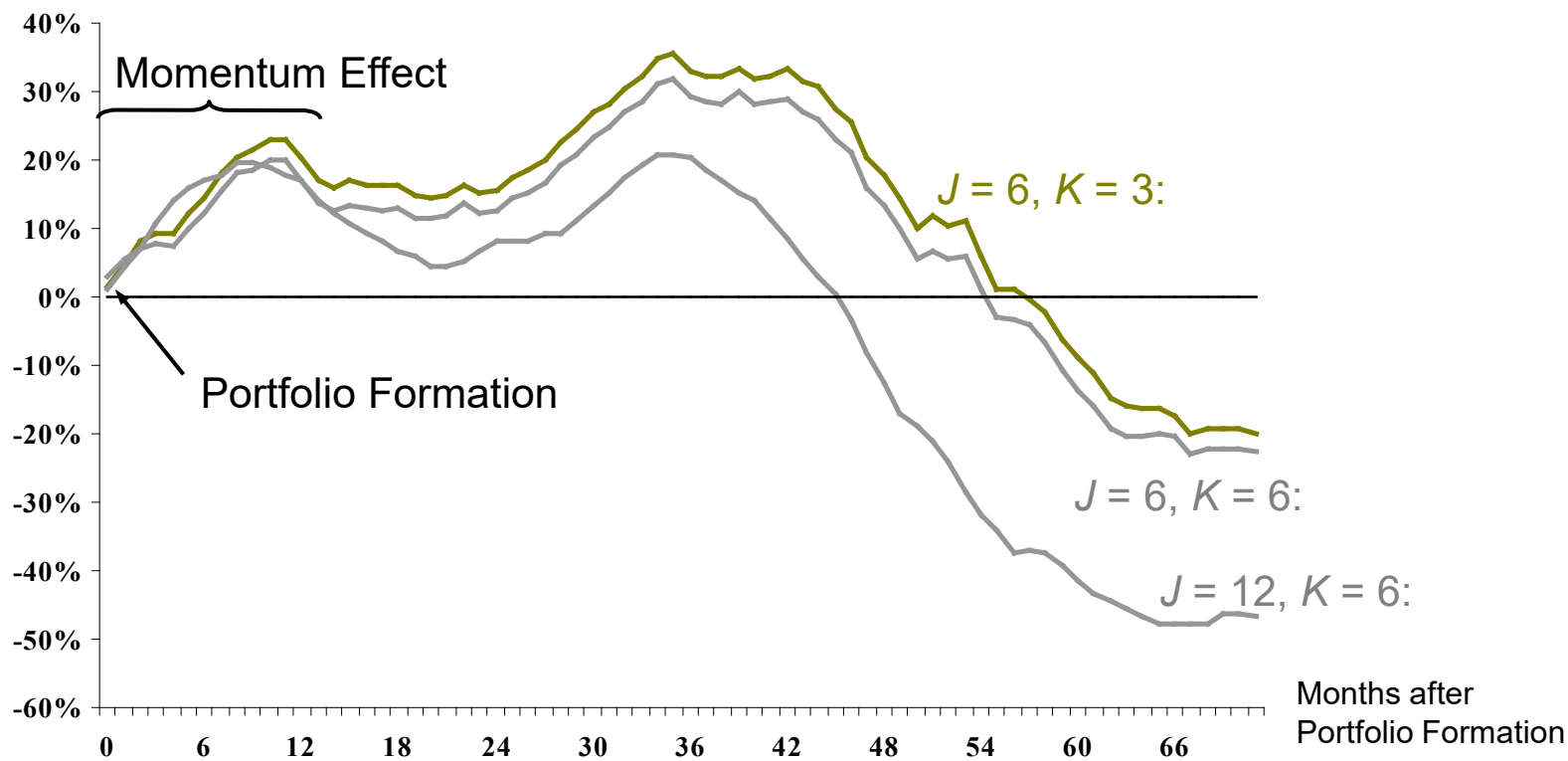
- One could expect overconfident people to trade more excessively since they are convinced about their superior investment ability.
- Psychological research has shown that men are more (over-) confident in their trading abilities, among other activities, than women.
- Do they trade more than women? How does their performance compare with that of women?
- Barber and Odean (2000) have obtained detailed account and trading data from over 30'000 households from a large discount brokerage firm in the U.S.

Behavioral finance: Individual trading behavior (II)

- Men trade significantly more than women: On average **45%**, single men even trade **67%** more than single women.
- The gross performance is about the same between men and women.
- However, the excessive trading of men hurts their performance due to high trading costs.
- Those individuals and mutual funds that trade the most, earn the lowest returns.
- Such empirical studies are powerful examples of overconfidence in human decision making.

Behavioral finance: Explanation for momentum

- The momentum pattern is consistent with underreaction to news and delayed overreaction (due to extrapolation): *Conservatism*
- Consistent: Large parts of momentum profits stem from the days around earnings announcements.



Behavioral corporate finance: eBay's valuation

- In 2003, 24 security analysts followed eBay. The most well-known was Mary Meeker from Morgan Stanley. She was one of the highest paid analysts at Wall Street having earned \$15m in 1999 and \$23m in 2000; she placed very high valuations on some internet firms, including eBay:

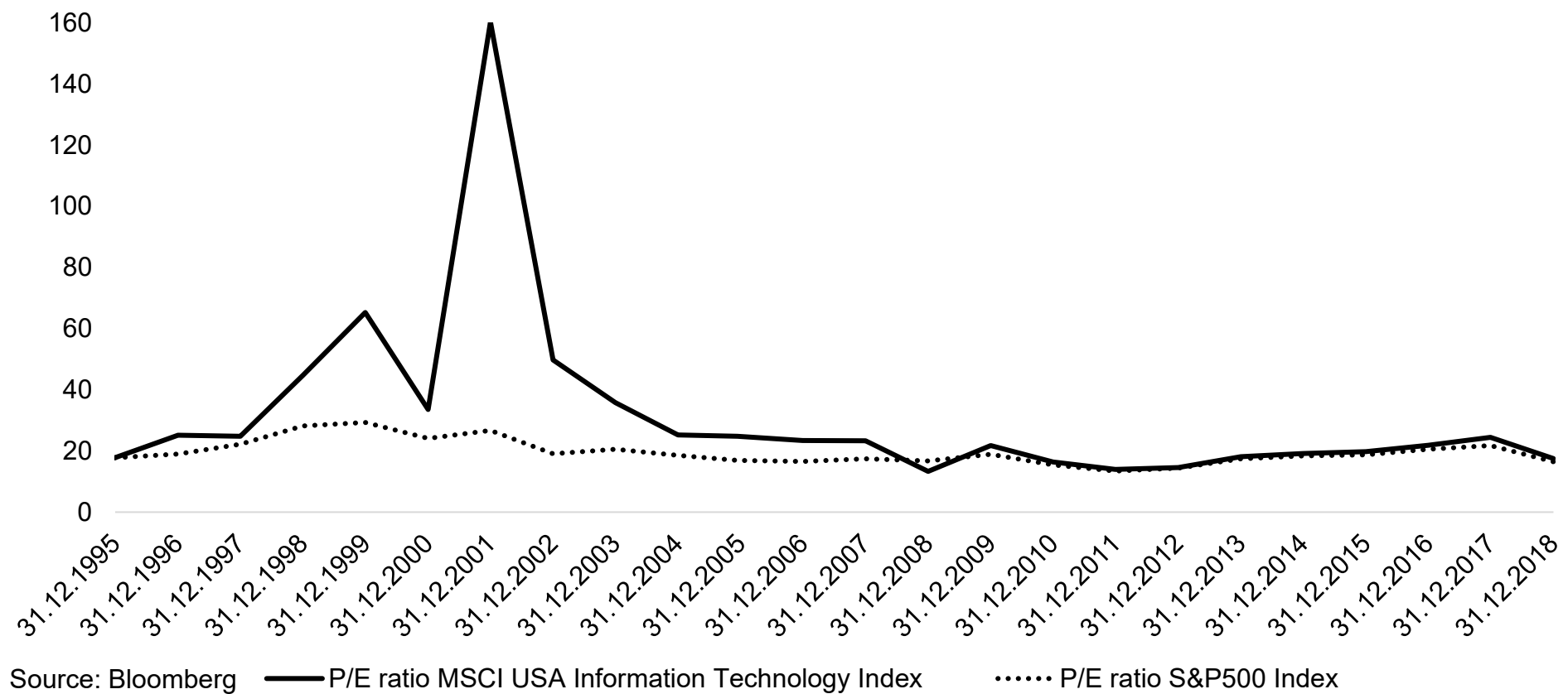
Technique	Downside	Upside	Base
P/E	\$74	\$111	\$84
PEG	\$45	\$91	\$68
Price-to-sales	\$97	\$210	\$147
Discounted cash flow			\$117
Average			\$104
Upside adjusted average			\$106

Source: Shefrin (2007, p. 25, Exhibit 2.1)

- In 2000, the price of many Internet stocks fell dramatically. The SEC investigated and criticized Mary Meeker's valuations, but did not put forward charges.
- Extrapolation bias: Another analyst admitted that his assumption that eBay could grow at a 30% compounded rate between 2002 and 2010 was unduly optimistic.

Behavioral corporate finance: eBay's valuation (II)

**Development of the P/E ratio of MSCI USA
Info Tech and the S&P500**



Behavioral corporate finance: CEO overconfidence

- Malmendier and Tate (2008) analyze the impact of CEO overconfidence on merger decisions.
- CEO overconfidence is measured by two proxies:
 - CEOs' personal over-investment in their company: Managers who fail to exercise in-the-money executive stock options prior to expiration.
 - CEO's press portrayal: CEOs who are characterized as "confident" or "optimistic" versus "cautious", "conservative", etc. by the business press.
- They find that the odds of making an acquisition are 65% higher if the CEO is classified as overconfident.
- The market reaction at merger announcement (-90 basis points) is significantly more negative than for non-overconfident CEOs (-12 basis points).
- ⇒ Overconfident CEOs overestimate their ability to generate returns. As a result, they overpay for target companies and undertake value-destroying mergers.
- ⇒ The effects are strongest if they have access to internal financing and thus are not subject to outside monitoring.

Key takeaways

1. There are **three forms of market efficiency**: weak, semi-strong, and strong.
2. There is a lot of empirical evidence which is consistent with at least weak and semi-strong form market efficiency and also a lot of evidence contradicting any of the three degrees of efficiency.
3. Still, markets are usually assumed to be **mostly semi-strong-form efficient**.
4. Behavioral finance questions the idea of rational human beings, and supports the conjecture of (partial) **irrationality** with vast empirical (often experimental) evidence. Inter alia, people show inconsistent attitudes about risk and are overconfident about the ability to make estimates about the future.

Literature

Mandatory readings

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Additional readings

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